

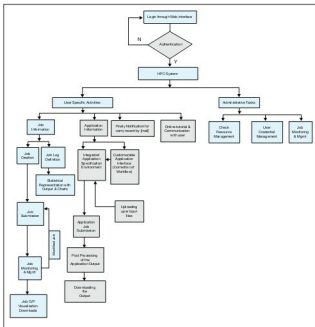


Figure 3: CHREME flow layout

- The Administrator Area

Salient features

The GUI-based Web interface: This organises, simplifies and modularises the access to various cluster resources through an intuitive graphical Web interface for users and administrators.

Comprehensive monitoring: Through CHREME, the HPC systems administrator can collect, monitor, visualise and analyse data pertaining to resource consumption, jobs, users, etc, thus empowering admins with actionable information.

Extensive runtime environment: This provides an extensive HPC runtime environment for parallel applications including compilers, scientific libraries, etc.

Security: Users access remote cluster resources from any location using a Web browser through HTTPS encryption and the SSH network protocol, which provides a secure data communication channel between the client machine and the HPC system.

Workload manager integration: CHREME is integrated with Torque, a resource manager, and with Maui, a cluster scheduler that helps users get optimal and efficient cluster resources.

Resource registry and management: The portal allows the creation, management and monitoring of various cluster resources like queues, parallel environments, users, nodes, etc.

Alerts and reporting: Users can receive automated notifications via e-mail, based on events such as job completion, changes in resource availability, job suspension, etc.

Credential management: This function is used to create new roles, new users, to assign/un-assign roles to users, and to disable/enable users from the HPC system. This is accessible only to the administrator.

Uploading and downloading files and folders: The portal offers a Web interface for file uploading and downloading from the client or HPC system. Errors and output files of the applications and jobs can be downloaded onto the client machine.

Benefits

CHREME provides an intuitive Web interface to ease job submissions, improve HPC system management and monitoring, and increase overall user productivity and cluster utilisation by integrating all its functions with the resource managers like Torque, PBSpro, etc.

- With CHREME, users do not have to learn Linux or the computational aspects of HPC, thus giving them the freedom to focus more on their core research area.
- CHREME usage is permeating to novice users, expert domain-specific scientists who are the end users, and to systems administrators.
- The intuitive GUI helps users submit their application-centric jobs and monitor the status of those jobs by receiving timely email notifications.
- CHREME provides systems administrators with HPC management and monitoring interfaces, thus making the systems management process a lot more easy and visual.
- Managements can better view, analyse and evaluate the work performed by the scientists and gauge efficient usage of the HPC infrastructure.
- CHREME's user-friendly interface to the HPC infrastructure that enables better visibility of the available resources, allows users to be more productive, leaving them with more time for innovation. This leads to reduced costs, improved product cycle times and increased efficiency of critical projects.

High Performance Computing (HPC) applications are common in many scientific domains. However, HPC is still complex and difficult to use for many researchers. CHREME makes it easier for scientists and researchers by abstracting the complex HPC environment, enabling them to use the HPC resources in an intuitive manner. Goldi Misra states that from the end user perspective, CHREME provides the convenience of accessing users' applications and corresponding data from a remote location, along with empowering the administrator with centralised management of all cluster resources. CHREME guides the user throughout the application execution process. With CHREME in place, the researcher is not burdened with the 'How' and 'Why' of running the HPC system. 

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The author revels in exploring new technologies and solutions. An avid reader of open source solutions, he has contributed code as well as documentation to open source projects and magazines. For him, open source is a philosophy and a way of life.